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Database Systems

Design, Implementation, & Management

National Economics University
Faculty of Mathematical Economics

Database System Design, Implementation And Management Course – Normalization of Database Tables

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Learning Objectives

- 6-1 Explain normalization and its role in the database design process
- 6-2 Identify each of the normal forms: 1NF, 2NF, 3NF, BCNF, 4NF, and 5NF
- 6-3 Explain how normal forms can be transformed from lower normal forms to higher normal forms
- 6-4 Apply normalization rules to evaluate and correct table structures
- 6-5 Identify situations that require denormalization to generate information efficiently
- 6-6 Use a data-modeling checklist to check that the ERD meets a set of minimum requirements

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6-1 Database Tables and Normalization

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- How do you recognize a poor table structure, and how do you produce a good table?
- **Normalization** is a process for evaluating and correcting table structures to minimize data redundancies.
- Involve assigning attributes to tables based on the concepts of determination and functional dependency.
- Normal forms: first three forms include first normal form (1NF), second normal form (2NF), and third normal form (3NF).
  - From a structural point of view, 2NF is better than 1NF, and 3NF is better than 2NF.
- For most purposes in business database design, 3NF is as high as you need to go in the normalization process

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6-1 Database Tables and Normalization

- The higher the normal form, the more relational join operations you need to produce a specified output. More resources are required by the database system to respond to end-user queries.
- **Denormalization** produces a lower normal form; that is, a 3NF will be converted to a 2NF through denormalization.
  - The price you pay for increased performance through denormalization is greater data redundancy

normalization
A process that assigns attributes to entities so that data redundancies are reduced or eliminated.

denormalization
A process by which a table is changed from a higher-level normal form to a lower-level normal form, usually to increase processing speed. Denormalization potentially yields data anomalies.

6-2 The Need for Normalization

6-2 The Need for Normalization

- Design process
  - Identifying business rules
  - Identifying and defining business and data constraints
  - Defining functional dependencies
  - Identifying entities and relationships
  - Eliminating multivalued attributes
- Design and normalize
  - Defining the business rules and data constraints, and identifying the functional dependencies, entities, and attributes
  - Apply normalization concepts to validate and further refine the model.

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Reporting activities of a construction

- ❑A construction company that manages several building projects.
- ❑Each project has its own project number, name, assigned employees, and so on.
- ❑Each employee has an employee number, name, and job classification, such as engineer or computer technician
- ❑The company charges its clients by billing the hours spent on each contract.
- ❑The hourly billing rate is dependent on the employee's job classification.
  - ➢One hour of computer technician time is billed at a different rate than one hour of engineer time.
- ❑A project report is generated that contains the information displayed in Table 6.1.

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Table 6.1 A Sample Report Layout

Project Number	Project Name	Employee Number	Employee Name	Job Class	Charge/Hour	Hours Billed	Total Charge
15	Evergreen	103	June E. Arbough	Elec. Engineer	$ 84.50	23.8	$ 2,011.10
		101	John G. News	Database Designer	$105.00	19.4	$ 2,037.00
		105	Alice K. Johnson *	Database Designer	$105.00	35.7	$ 3,748.50
		106	William Smithfield	Programmer	$ 35.75	12.6	$ 450.45
		102	David H. Senior	Systems Analyst	$ 96.75	23.8	$ 2,302.65
				Subtotal			**$10,549.70**
18	Amber Wave	114	Annelise Jones	Applications Designer	$ 48.10	24.6	$ 1,183.26
		118	James J. Frommer	General Support	$ 18.36	45.3	$ 831.71
		104	Anne K. Ramoras *	Systems Analyst	$ 96.75	32.4	$ 3,134.70
		112	Darlene M. Smithson	DSS Analyst	$ 45.95	44.0	$ 2,021.80
				Subtotal			**$ 7,171.47**
22	Rolling Tide	105	Alice K. Johnson	Database Designer	$105.00	64.7	$ 6,793.50
		104	Anne K. Ramoras	Systems Analyst	$ 96.75	48.4	$ 4,682.70
		113	Delbert K. Joenbrood *	Applications Designer	$ 48.10	23.6	$ 1,135.16
		111	Geoff B. Wabash	Clerical Support	$ 26.87	22.0	$ 591.14
		106	William Smithfield	Programmer	$ 35.75	12.8	$ 457.60
				Subtotal			**$13,660.10**
25	Starflight	107	Maria D. Alonzo	Programmer	$ 35.75	24.6	$ 879.45
		115	Travis B. Bawangi	Systems Analyst	$ 96.75	45.8	$ 4,431.15
		101	John G. News *	Database Designer	$105.00	56.3	$ 5,911.50
		114	Annelise Jones	Applications Designer	$ 48.10	33.1	$ 1,592.11
		108	Ralph B. Washington	Systems Analyst	$ 96.75	23.6	$ 2,283.30
		118	James J. Frommer	General Support	$ 18.36	30.5	$ 559.98
		112	Darlene M. Smithson	DSS Analyst	$ 45.95	41.4	$ 1,902.33
				Subtotal			**$17,559.82**
				Total			**$48,941.09**

Note: * indicates the project leader.

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Table name: RPT_FORMAT **Database name: Ch06_ConstructCo**

PROJ_NUM	PROJECT_NAME	EMP_NUMBER	EMP_NAME	JOB_CLASS	CHARGE_HOUR	HOURS_BILLED
15	Evergreen	103,101,105, 106, 102	June E. Arbough, John G. News, Alice K. Johnson *, William Smithfield, David H. Senior	Elec. Engineer, Database Designer, Database Designer, Programmer, Systems Analyst	85.5, 105., 105., 35.75, 98.75	23.8, 19.4, 35.7, 12.6, 23.8
18	Amber Wave	114, 118, 104, 112	Annelise Jones, James J. Frommer, Anne K. Ramoras *, Darlene M. Smithson	Applications Designer, General Support, Systems Analyst, DSS Analyst	48.1, 18.36, 96.75, 45.95	25.6, 45.3, 32.4, 45.
22	Rolling Tide	105, 104, 113, 111, 106	Alice K. Johnson, Anne K. Ramoras, Delbert K. Joenbrood *, Geoff B. Wabash, William Smithfield	DB Designer, Systems Analyst, Applications Designer, Clerical Support, Programmer	105., 96.75, 48.1, 26.87, 35.75	65.7, 48.4, 23.6, 22., 12.8
25	Starflight	107, 115, 101, 114, 108, 118, 112	Maria D. Alonzo, Travis B. Bawangi, John G. News *, Annelise Jones, Ralph B. Washington, James J. Frommer, Darlene M. Smithson	Programmer, Systems Analyst, Database Design, Applications Designer, Systems Analyst, General Support, DSS Analyst	35.75, 96.75, 105., 48.1, 96.75, 18.36, 45.95	25.6, 45.8, 56.3, 33.1, 23.6, 30.5, 41.4

unnormalized data, reflected by the existence of several multivalued data elements (EMP_NUM, EMP_NAME, JOB_CLASS, CHARGE_HOUR, HOURS_BILLED)

Deficiencies

- Data inconsistencies.
  - “Elect. Engineer” might be entered as “Elect.Eng.”, “El. Eng.” and “EE”.
  - John G. News and Alice K. Johnson in the Evergreen project to charge different rates even though they have the same job classification.
- contains several multivalued attributes
  - hard to identify each employee individually and for the database to answer questions such as “How many employees are working on the Starflight project
- Employee data is redundant

6-3 The Normalization Process

6-3 The Normalization Process

- Concept of well-formed relations:
  - Each relation (table) represents a single subject
  - Each row/column intersection contains only one (a single) value and not a group of values.
  - No data item will be *unnecessarily* stored in more than one table
  - All nonprime attributes in a relation (table) are dependent on the primary key—the entire primary key and nothing but the primary key
  - Each relation (table) has no insertion, update, or deletion anomalies, which ensures the integrity and consistency of the data.

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Table 6.3 Functional Dependence Concepts

Concept	Definition
Functional dependence	The attribute *B* is fully functionally dependent on the attribute *A* if each value of *A* determines one and only one value of *B*. Example: PROJ_NUM → PROJ_NAME (read as *PROJ_NUM functionally determines PROJ_NAME*) In this case, the attribute PROJ_NUM is known as the determinant attribute, and the attribute PROJ_NAME is known as the dependent attribute.
Functional dependence (generalized definition)	Attribute *A* determines attribute *B* (that is, *B* is functionally dependent on *A*) if all of the rows in the table that agree in value for attribute *A* also agree in value for attribute *B*.
Fully functional dependence (composite key)	If attribute *B* is functionally dependent on a composite key *A* but not on any subset of that composite key, the attribute *B* is fully functionally dependent on *A*.

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Functional Dependence Concepts

- **partial dependency:** A condition in which an attribute is dependent on only a portion (subset) of the primary key.
  - If (A, B) → (C, D), B → C, and (A, B) is the primary key, then the functional dependence B → C is a partial dependency because only part of the primary key (B) is needed to determine the value of C.
- **transitive dependency:** A condition in which an attribute is dependent on another attribute that is not part of the primary key
  - There are functional dependencies such that X → Y, Y → Z, and X is the primary key. In that case, the dependency X → Z is a transitive dependency because X determines the value of Z via Y

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Table 6.2 Normal Forms

Normal Form	Characteristic	Section
First normal form (1NF)	Table format, no repeating groups, and PK identified	6-3a
Second normal form (2NF)	1NF and no partial dependencies	6-3b
Third normal form (3NF)	2NF and no transitive dependencies	6-3c
Boyce-Codd normal form (BCNF)	3NF and every determinant is a candidate key (special case of 3NF)	6-6a
Fourth normal form (4NF)	BCNF and no independent multivalued dependencies	6-6b
Fifth normal form (5NF or PJNF)	4NF and cannot have lossless decomposition into smaller tables	6-6c

Note

The term **first normal form (1NF)** describes the tabular format that conforms to the definition of a relational table in which:

- All of the key attributes are defined.
- There are no repeating groups in the table. In other words, each row/column intersection contains one and only one value, not a set of values.
- All attributes are dependent on the primary key.

6-3a Conversion to First Normal Form (1NF)

- A **repeating group** derives its name from the fact that a group of multiple entries of the same or multiple types can exist for any *single* key attribute occurrence.
  - PROJ_NUM occurrence can reference a group of related data in the employee number, employee name, job classification, and charge per hour columns.
  - Evergreen project (PROJ_NUM = 15) contains five values for each of those attributes
- Eliminate repeating group by making sure that each row defines a single entity instance and that each row-column intersection has only a single value.
- The dependencies must be identified to diagnose the normal form.
- Normalization starts with a simple three-step procedure

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6-3a Conversion to First Normal Form (1NF)

Step 1: Eliminate the Repeating Groups

- Start by presenting the data in a tabular format, where each cell has a single value and there are no repeating groups.
- To eliminate the repeating groups, change the table from a project focus to an assignment focus.
- This will create separate rows for each employee assigned to each project, converting the multivalued attributes into single-valued attributes.
- This change converts the table in Figure 6.1 to 1NF as shown in Figure 6.2.

Figure 6.2 A Table in First Normal Form

Table name: DATA_ORG_1NF **Database name: Ch06_ConstructCo**

PROJ_NUM	PROJ_NAME	EMP_NUM	EMP_NAME	JOB_CLASS	CHG_HOUR	HOURS
15	Evergreen	103	June E. Arbough	Elect. Engineer	84.50	23.8
15	Evergreen	101	John G. News	Database Designer	105.00	19.4
15	Evergreen	105	Alice K. Johnson *	Database Designer	105.00	35.7
15	Evergreen	106	William Smithfield	Programmer	35.75	12.6
15	Evergreen	102	David H. Senior	Systems Analyst	96.75	23.8
18	Amber Wave	114	Annelise Jones	Applications Designer	48.10	24.6
18	Amber Wave	118	James J. Frommer	General Support	18.36	45.3
18	Amber Wave	104	Anne K. Ramoras *	Systems Analyst	96.75	32.4
18	Amber Wave	112	Darlene M. Smithson	DSS Analyst	45.95	44.0
22	Rolling Tide	105	Alice K. Johnson	Database Designer	105.00	64.7
22	Rolling Tide	104	Anne K. Ramoras	Systems Analyst	96.75	48.4
22	Rolling Tide	113	Delbert K. Joenbrood *	Applications Designer	48.10	23.6
22	Rolling Tide	111	Geoff B. Wabash	Clerical Support	26.87	22.0
22	Rolling Tide	106	William Smithfield	Programmer	35.75	12.8
25	Starflight	107	Maria D. Alonzo	Programmer	35.75	24.6
25	Starflight	115	Travis B. Bawangi	Systems Analyst	96.75	45.8
25	Starflight	101	John G. News *	Database Designer	105.00	56.3
25	Starflight	114	Annelise Jones	Applications Designer	48.10	33.1
25	Starflight	108	Ralph B. Washington	Systems Analyst	96.75	23.6
25	Starflight	118	James J. Frommer	General Support	18.36	30.5
25	Starflight	112	Darlene M. Smithson	DSS Analyst	45.95	41.4

6-3a Conversion to First Normal Form (1NF)

❑Step 2: Identify the Primary Key

- PROJ_NUM is not an adequate primary key because the project number does not uniquely identify each row.
- To maintain a proper primary key that will *uniquely* identify any attribute value, the new key must be composed of a *combination* of PROJ_NUM and EMP_NUM.
  - ✓PROJ_NUM = 15 and EMP_NUM = 103, the entries for the attributes PROJ_NAME, EMP_NAME, JOB_CLASS, CHG_HOUR, and HOURS must be Evergreen, June E. Arbough, Elect. Engineer, $84.50, and 23.8, respectively.

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6-3a Conversion to First Normal Form (1NF)

- Step 3: Identify All Dependencies
  - The identification of the PK in Step 2 means that you have already identified the following dependency:
    PROJ_NUM, EMP_NUM → PROJ_NAME, EMP_NAME, JOB_CLASS, CHG_HOUR, HOURS
  - The PROJ_NAME, EMP_NAME, JOB_CLASS, CHG_HOUR, and HOURS values are all dependent on—they are determined by—the combination of PROJ_NUM and EMP_NUM.

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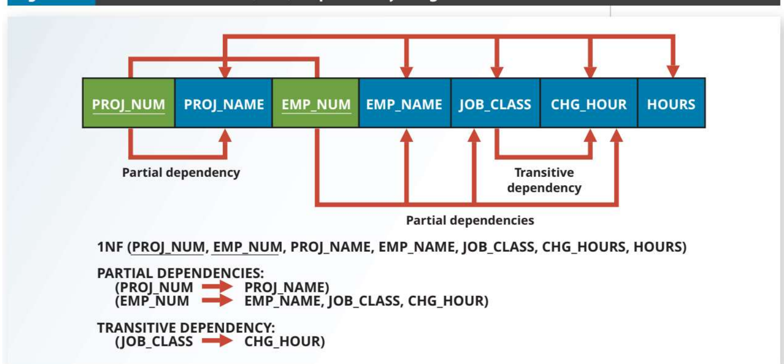
6-3a Conversion to First Normal Form (1NF)

- Anomalies remain
- Each time another employee is assigned to a project, some data entries (such as PROJ_NAME, EMP_NAME, and CHG_HOUR) are unnecessarily repeated
- The anomalies that remain exist because there are additional dependencies in addition to the primary key dependency.
  - Project number determines the project name: PROJ_NUM → PROJ_NAME
  - Employee number determines employee’s name, job classification, and charge per hour: EMP_NUM → EMP_NAME, JOB_CLASS, CHG_HOUR

Dependency diagram

- A representation of all data dependencies (primary key, partial, or transitive) within a table.

Figure 6.3 First Normal Form (1NF) Dependency Diagram



Anomalies

- *Update anomalies*. Modifying the JOB_CLASS for an employee requires updating many entries; otherwise, it will generate data inconsistencies.
- *Insertion anomalies*. Adding a new employee requires the employee to be assigned to a project and therefore to enter duplicate project information. If the employee is not yet assigned to a project, a phantom project must be created to complete the employee data entry.
- *Deletion anomalies*. Suppose that only one employee is associated with a given project. If that employee is deleted, the project information will also be deleted.

6-3b Conversion to Second Normal Form (2NF)

❑Conversion to 2NF occurs only when the 1NF has a composite primary key.

.

Note

A table is in **second normal form (2NF)** when:

- It is in 1NF

and

- It includes no partial dependencies; that is, no attribute is dependent on only a portion of the primary key.

It is still possible for a table in 2NF to exhibit transitive dependency. That is, the primary key may rely on one or more nonprime attributes to functionally determine other nonprime attributes, as indicated by a functional dependence among the nonprime attributes.

6-3b Conversion to Second Normal Form (2NF)

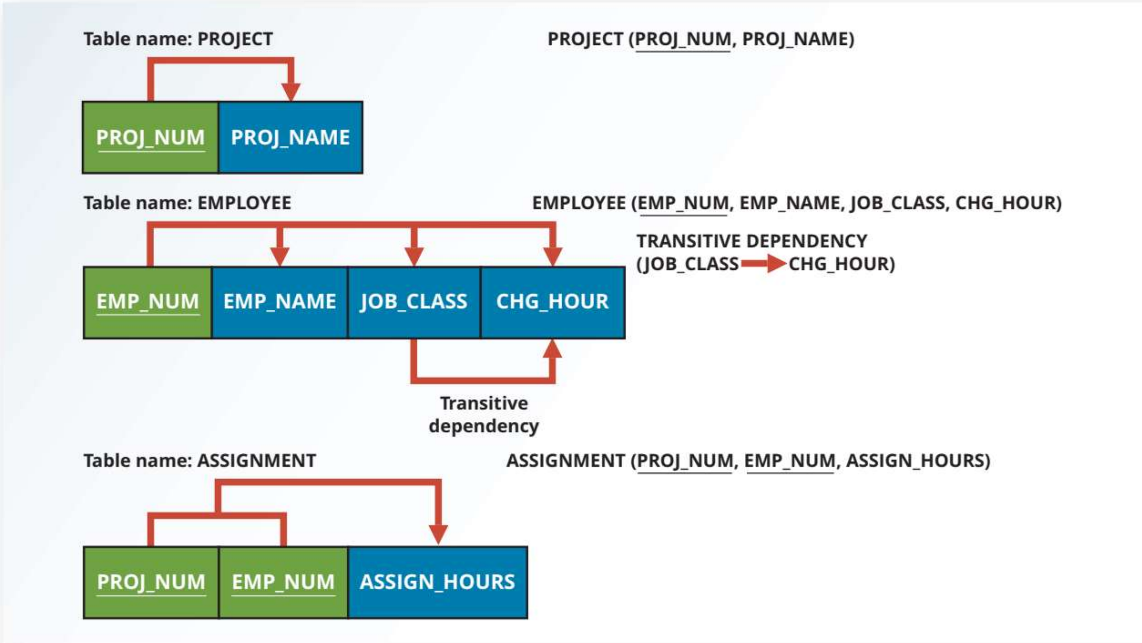
❑Step 1: Make New Tables to Eliminate Partial Dependencies For each component of the primary key that acts as a determinant in a partial dependency, create a new table with a copy of that component as the primary key.

- The original table is now divided into three tables (PROJECT, EMPLOYEE, and ASSIGNMENT).

❑Step 2: Reassign Corresponding Dependent Attributes

- PROJECT (**PROJ_NUM**, PROJ_NAME)
  EMPLOYEE (**EMP_NUM**, EMP_NAME, JOB_CLASS, CHG_HOUR)
  ASSIGNMENT (**PROJ_NUM, EMP_NUM**, ASSIGN_HOURS)

Figure 6.4 Second Normal Form (2NF) Conversion Results



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6-3c Conversion to Third Normal Form (3NF)

- A transitive dependency can generate anomalies.
- **third normal form (3NF)** A table is in 3NF when it is in 2NF and no nonkey attribute is functionally dependent on another nonkey attribute; that is, it cannot include transitive dependencies.

> **Note**
>
> A table is in **third normal form (3NF)** when:
>
> - It is in 2NF
>
> *and*
>
> - It contains no transitive dependencies.

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6-3c Conversion to Third Normal Form (3NF)

- Step 1: Make New Tables to Eliminate Transitive Dependencies
  - For every transitive dependency, write a copy of its determinant (JOB_CLASS) as a primary key for a new table. the determinant remain in the original table to serve as a foreign key
- Step 2: Reassign Corresponding Dependent Attributes identify the attributes that are dependent on each determinant identified in Step 1. Place the dependent attributes in the new tables with their determinants and remove them from their original tables.
  - In this example, eliminate CHG_HOUR from the EMPLOYEE table shown in Figure 6.4 to leave the EMPLOYEE table dependency definition as: EMP_NUM → EMP_NAME, JOB_CLASS

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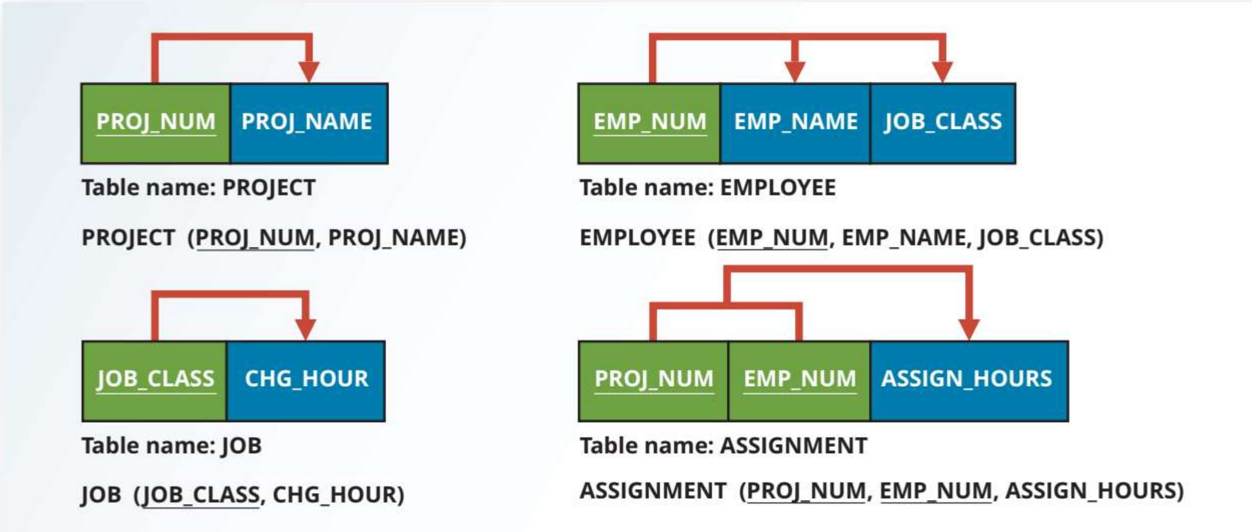
6-3c Conversion to Third Normal Form (3NF)

- A transitive dependency can generate anomalies.
- Step 1: Make New Tables to Eliminate Transitive Dependencies
  - For every transitive dependency, write a copy of its determinant (JOB_CLASS) as a primary key for a new table. the determinant remain in the original table to serve as a foreign key
- Step 2: Reassign Corresponding Dependent Attributes identify the attributes that are dependent on each determinant identified in Step 1. Place the dependent attributes in the new tables with their determinants and remove them from their original tables.
  - Eliminate CHG_HOUR from the EMPLOYEE table shown in Figure 6.4 to leave the EMPLOYEE table dependency definition as: EMP_NUM → EMP_NAME, JOB_CLASS

Figure 6.5 Third Normal Form (3NF) Conversion Results



6-6 Higher-Level Normal Forms

6-6a The Boyce-Codd Normal Form

- **Boyce-Codd normal form (BCNF)** A special type of third normal form (3NF) in which every determinant is a candidate key. A table in BCNF must be in 3NF.
  - A candidate key has the same characteristics as a primary key, but for some reason, it was not chosen to be the primary key
  - BCNF can be violated only when the table contains more than one candidate key

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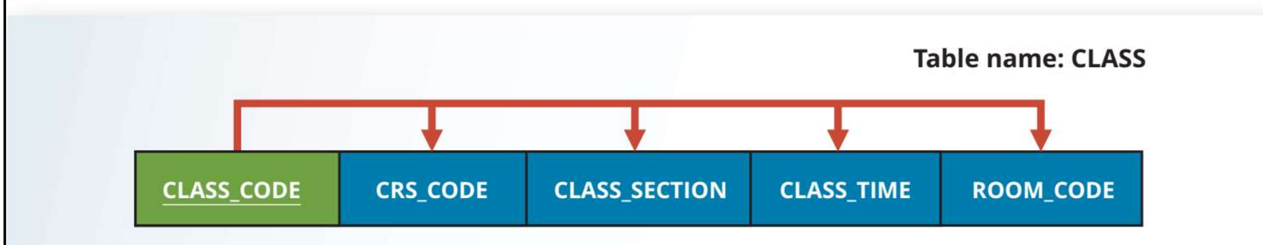
Figure 6.7 Tables with Multiple Candidate Keys



The CLASS table has two candidate keys:
- CLASS_CODE
- CRS_CODE + CLASS_SECTION

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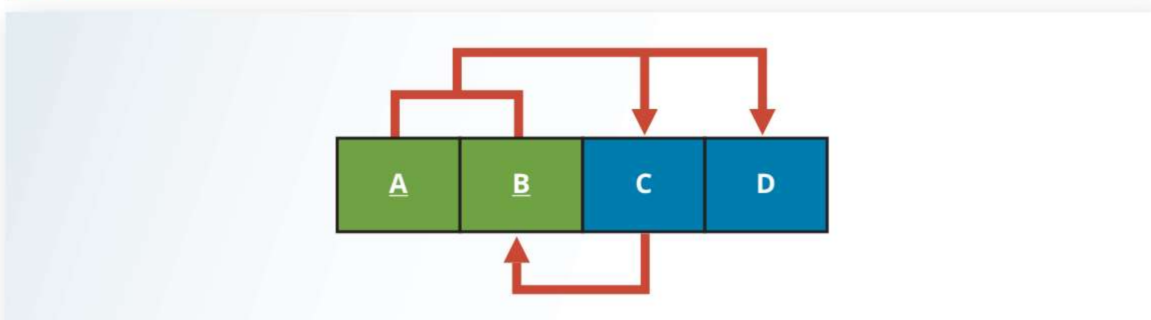
6-6a The Boyce-Codd Normal Form

- The table is in 1NF because the key attributes are defined and all nonkey attributes are determined by the key.
- The table is in 2NF because it is in 1NF and there are no partial dependencies on either candidate key.
- The table is in 3NF because there are no transitive dependencies.
- A table be in 3NF and not be in BCNF?
  - one key attribute is the determinant of another key attribute
  - because BCNF requires that every determinant in the table be a candidate key

Figure 6.8 A Table That Is in 3NF but not in BCNF



The functional dependencies in Figure 6.8:

A + B → C, D

A + C → B, D

C → B : *one key attribute determines part of the primary key*

The condition C → B causes the table to fail to meet the BCNF requirements.

An attribute that is a part of a key is called a **key attribute**

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Figure 6.9 Decomposition to BCNF

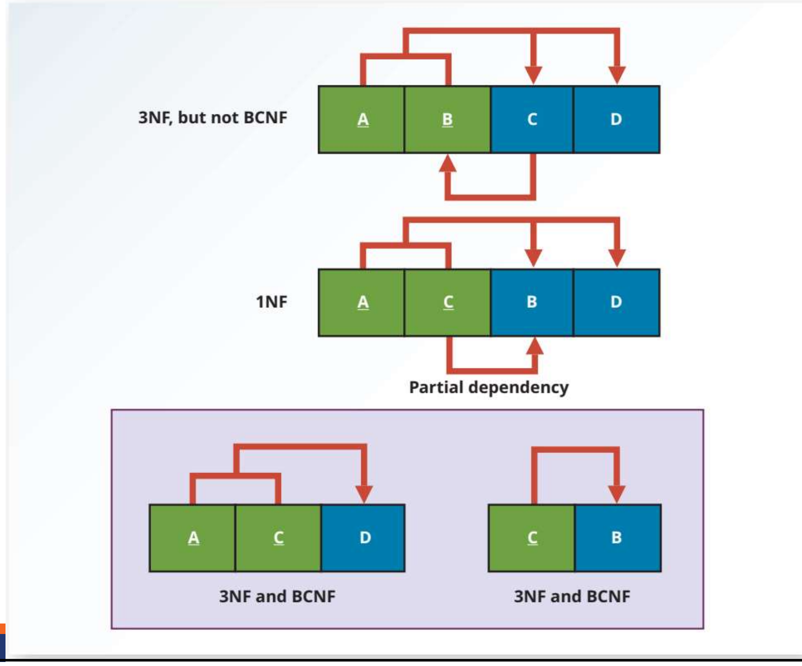

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Table 6.5 Sample Data for a BCNF Conversion

Stu_ID	Staff_ID	Class_code	Enroll_grade
125	25	21334	A
125	20	32456	C
135	20	28458	B
144	25	27563	C
144	20	32456	B

The structure shown in Table 6.5 is reflected in Panel A of Figure 6.10:
STU_ID + STAFF_ID → CLASS_CODE, ENROLL_GRADE
CLASS_CODE → STAFF_ID

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6-6a The Boyce-Codd Normal Form

- Each CLASS_CODE identifies a class uniquely
  - Illustrate the case in which a course might generate many classes.
- A student can take many classes.
- A staff member can teach many classes, but each class is taught by only one staff member
- The structure shown in Table 6.5 is reflected in Panel A of Figure 6.10:

  STU_ID + STAFF_ID → CLASS_CODE, ENROLL_GRADE

  CLASS_CODE → STAFF_ID

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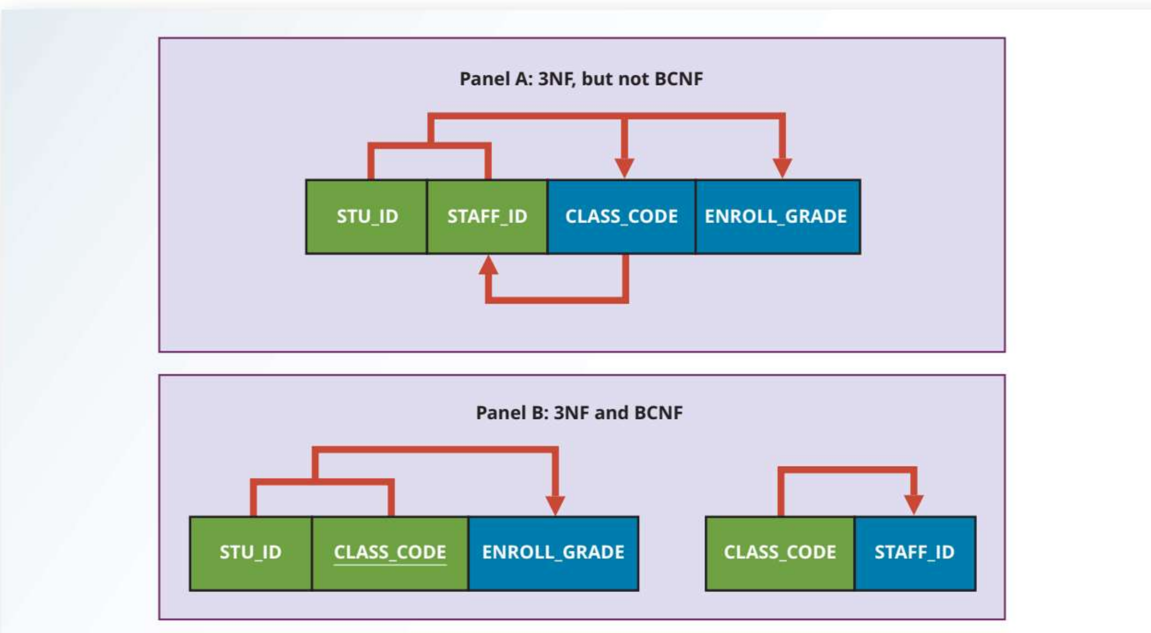
Figure 6.10 Another BCNF Decomposition



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6-6b Fourth Normal Form (4NF)

- **fourth normal form (4NF)** A table is in 4NF if it is in BCNF and contains no multiple independent sets of multivalued dependencies.
- Creating new tables for the components of the multivalued dependency.

Note
A table is in **fourth normal form (4NF)** when it is in 3NF and has no multivalued dependencies.

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Figure 6.11 Tables with Multivalued Dependencies

Database name: Ch06_Service

Table name: VOLUNTEER_V1

EMP_NUM	ORG_CODE	ASSIGN_NUM
10123	RC	1
10123	UW	3
10123		4

Table name: VOLUNTEER_V2

EMP_NUM	ORG_CODE	ASSIGN_NUM
10123	RC	
10123	UW	
10123		1
10123		3
10123		4

Table name: VOLUNTEER_V3

EMP_NUM	ORG_CODE	ASSIGN_NUM
10123	RC	1
10123	RC	3
10123	UW	4

An employee can have multiple assignments and can also be involved in multiple service organizations. Employee 10123 volunteers for the Red Cross and United Way. In addition, the same employee might be assigned to work on three projects: 1, 3, and 4.
Figure 6.11 illustrates how that set of facts can be recorded in very different ways.

Figure 6.12 A Set of Tables in 4NF

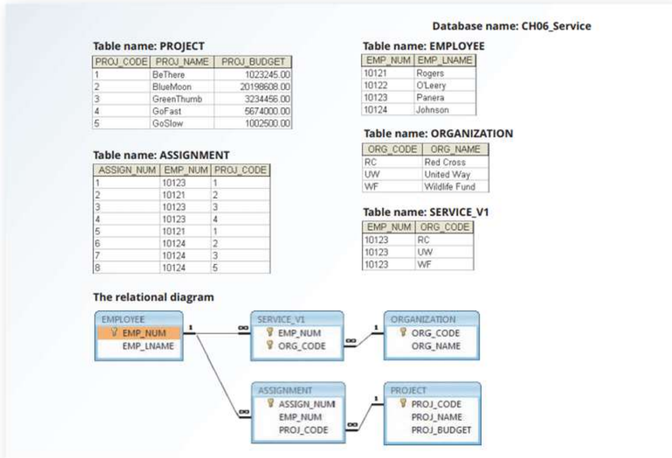

Table name: PROJECT

PROJ_CODE	PROJ_NAME	PROJ_BUDGET
1	BeThere	1023245.00
2	BlueMoon	20198608.00
3	GreenThumb	3234456.00
4	GoFast	5674000.00
5	GoSlow	1002500.00

Table name: EMPLOYEE

EMP_NUM	EMP_LNAME
10121	Rogers
10122	O'Leery
10123	Panera
10124	Johnson

Table name: ASSIGNMENT

ASSIGN_NUM	EMP_NUM	PROJ_CODE
1	10123	1
2	10121	2
3	10123	3
4	10123	4
5	10121	1
6	10124	2
7	10124	3
8	10124	5

Table name: ORGANIZATION

ORG_CODE	ORG_NAME
RC	Red Cross
UW	United Way
WF	Wildlife Fund

Table name: SERVICE_V1

EMP_NUM	ORG_CODE
10123	RC
10123	UW
10123	WF

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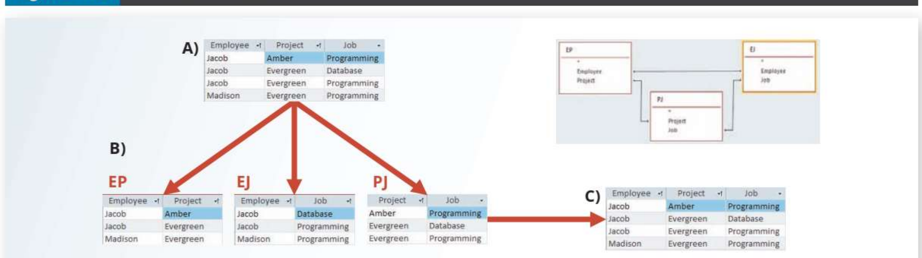
6-6c Fifth Normal Form (5NF)

- Fifth normal form, also known as **project join normal form (PJNF)**, addresses the issue in which a table cannot be decomposed anymore without losing data or creating incorrect information
- **project join normal form (PJNF)** Another term for fifth normal form (5NF). A state in which a table is already in 4NF and has no lossless decompositions

Figure 6.13 5NF Conversion



A)

Employee	Project	Job
Jacob	Amber	Programming
Jacob	Evergreen	Database
Jacob	Evergreen	Programming
Madison	Evergreen	Programming

B) EP

Employee	Project
Jacob	Amber
Jacob	Evergreen
Madison	Evergreen

EJ

Employee	Job
Jacob	Database
Jacob	Programming
Madison	Programming

PJ

Project	Job
Amber	Programming
Evergreen	Database
Evergreen	Programming

C)

Employee	Project	Job
Jacob	Amber	Programming
Jacob	Evergreen	Database
Jacob	Evergreen	Programming
Madison	Evergreen	Programming

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6-7 Normalization and Database Design

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6-7 Normalization and Database Design

- Normalization should be part of the design process.
- Make sure that proposed entities meet the required normal form before the table structures are created
- Second, normalization focuses on the characteristics of specific entities; that is, normalization represents a micro view of the entities within the ERD

Operations of the contracting company

- The company manages many projects.
- Each project requires the services of many employees.
- An employee may be assigned to several different projects.
- Some employees are not assigned to a project and perform duties not specifically related to a project. Some employees are part of a labor pool, to be shared by all project teams. For example, the company's executive secretary would not be assigned to any one particular project.
- Each employee has a single primary job classification, which determines the hourly billing rate.
- Many employees can have the same job classification. For example, the company employs more than one electrical engineer.

Given that simple description of the company's operations, two entities and their attributes are initially defined:

- PROJECT (**PROJ_NUM**, PROJ_NAME)
- EMPLOYEE (**EMP_NUM**, EMP_LNAME, EMP_FNAME, EMP_INITIAL, JOB_DESCRIPTION, JOB_CHG_HOUR)

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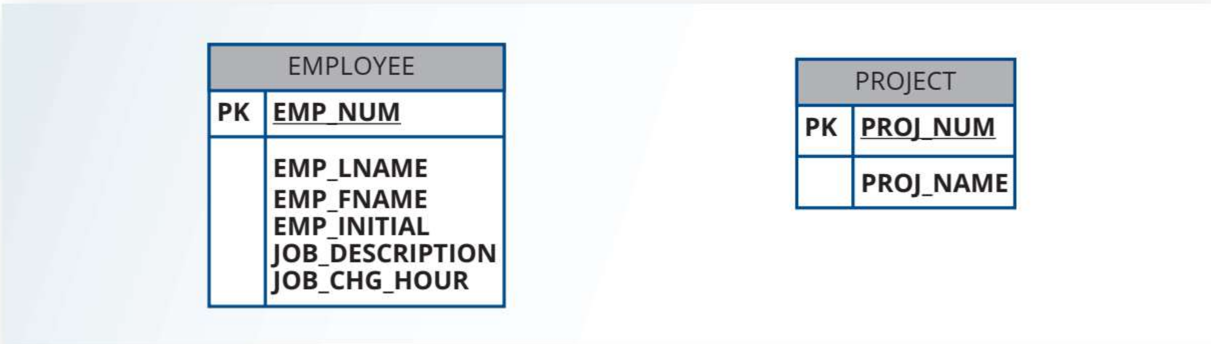
Figure 6.14 Initial Contracting Company ERD



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After creating the initial ERD shown in Figure 6.14, the normal forms are defined:

- PROJECT is in 3NF and needs no modification at this point.
- EMPLOYEE requires additional scrutiny. The JOB_DESCRIPTION attribute defines job classifications such as Systems Analyst, Database Designer, and Programmer. In turn, those classifications determine the billing rate, JOB_CHG_HOUR. Therefore, EMPLOYEE contains a transitive dependency.

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The removal of EMPLOYEE's transitive dependency yields three entities:

- PROJECT (**PROJ_NUM**, PROJ_NAME)
- EMPLOYEE (**EMP_NUM**, EMP_LNAME, EMP_FNAME, EMP_INITIAL, JOB_CODE)
- JOB (**JOB_CODE**, JOB_DESCRIPTION, JOB_CHG_HOUR)

Figure 6.15 Modified Contracting Company ERD

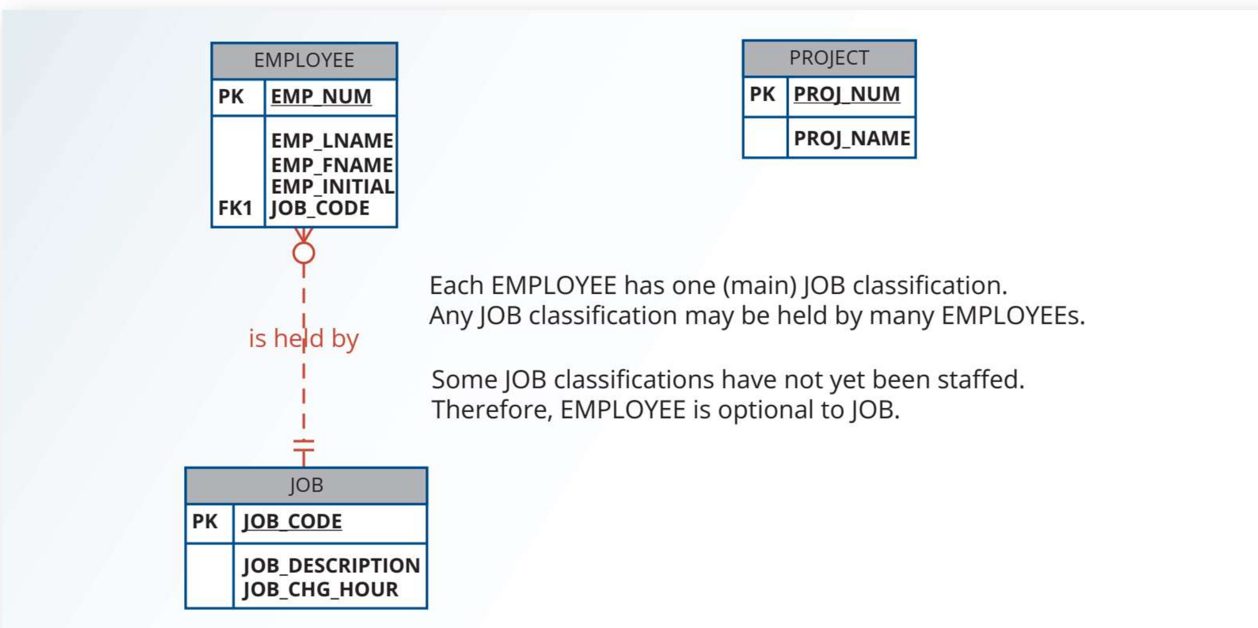

Figure 6.16 Incorrect M:N Relationship Representation

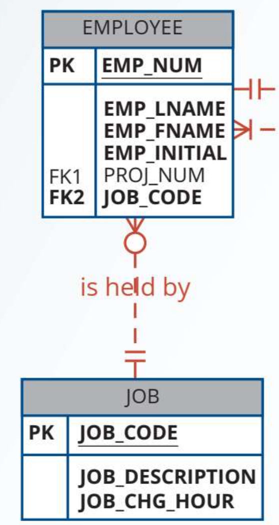

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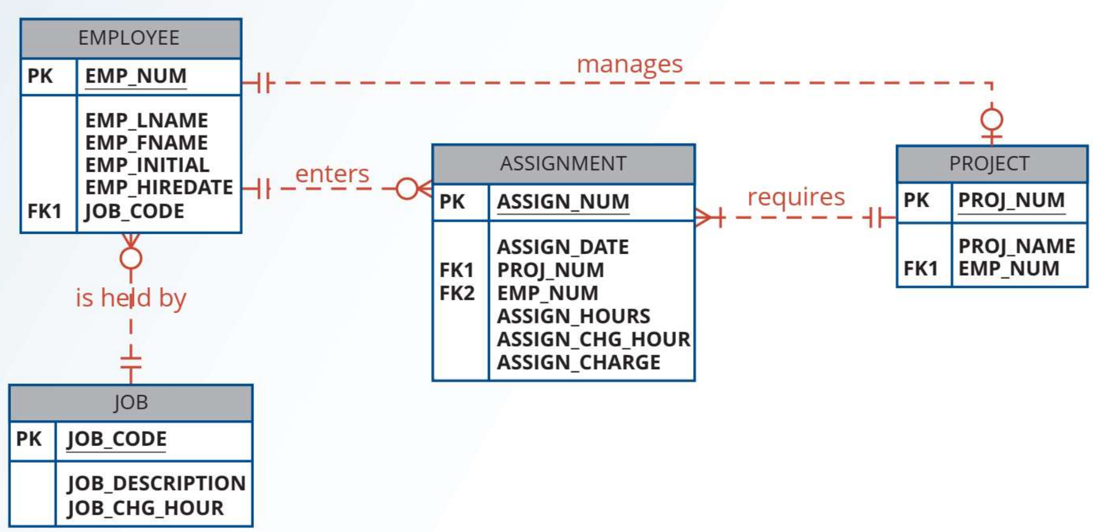
Figure 6.17 Final Contracting Company ERD



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PROJECT (**PROJ_NUM**, PROJ_NAME, EMP_NUM)

EMPLOYEE (**EMP_NUM**, EMP_LNAME, EMP_FNAME, EMP_INITIAL, EMP_HIREDATE, JOB_CODE)

JOB (**JOB_CODE**, JOB_DESCRIPTION, JOB_CHG_HOUR)

ASSIGNMENT (**ASSIGN_NUM**, ASSIGN_DATE, PROJ_NUM, EMP_NUM, ASSIGN_HOURS, ASSIGN_CHG_HOUR, ASSIGN_CHARGE)

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6-8 Denormalization

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Table 6.6 Common Denormalization Examples

Case	Example	Rationale and Controls
Redundant data	Storing ZIP and CITY attributes in the AGENT table when ZIP determines CITY (see Figure 2.2)	Avoid extra join operations Program can validate city (drop-down box) based on the zip code
Derived data	Storing STU_HRS and STU_CLASS (student classification) when STU_HRS determines STU_CLASS (see Figure 3.28)	Avoid extra join operations Program can validate classification (lookup) based on the student hours
Preaggregated data (also derived data)	Storing the student grade point average (STU_GPA) aggregate value in the STUDENT table when this can be calculated from the ENROLL and COURSE tables (see Figure 3.28)	Avoid extra join operations Program computes the GPA every time a grade is entered or updated STU_GPA can be updated only via administrative routine
Information requirements	Using a temporary denormalized table to hold report data; this is required when creating a tabular report in which the columns represent data that are stored in the table as rows (see Figures 6.18 and 6.19)	Impossible to generate the data required by the report using plain SQL No need to maintain table Temporary table is deleted once report is done Processing speed is not an issue

6-9 Data-Modeling Checklist

Table 6.7 Data-Modeling Checklist

Business Rules

- Properly document and verify all business rules with the end users.
- Ensure that all business rules are written precisely, clearly, and simply. The business rules must help identify entities, attributes, relationships, and constraints.
- Identify the source of all business rules, and ensure that each business rule is justified, dated, and signed off by an approving authority.

Data Modeling

Naming conventions: All names should be limited in length (database-dependent size).

- Entity names:
  - Should be nouns that are familiar to business and should be short and meaningful.
  - Should document abbreviations, synonyms, and aliases for each entity.
  - Should be unique within the model.
  - Composite entities may include a combination of abbreviated names of the entities linked through the composite entity.
- Attribute names:
  - Should be unique within the entity.
  - Should use the entity abbreviation as a prefix.
  - Should be descriptive of the characteristic.
  - Should use suffixes such as _ID, _NUM, or _CODE for the PK attribute.
  - Should not be a reserved word.
  - Should not contain spaces or special characters such as @, !, or &.
- Relationship names:
  - Should be active or passive verbs that clearly indicate the nature of the relationship.

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Entities:

- Each entity should represent a single subject.
- Each entity should represent a set of distinguishable entity instances.
- All entities should be in 3NF or higher. Any entities below 3NF should be justified.
- The granularity of the entity instance should be clearly defined.
- The PK should be clearly defined and support the selected data granularity.

Attributes:

- Should be simple and single-valued (atomic data).
- Should document default values, constraints, synonyms, and aliases.
- Derived attributes should be clearly identified and include source(s).
- Should not be redundant unless this is required for transaction accuracy, performance, or maintaining a history.
- Nonkey attributes must be fully dependent on the PK attribute.

Relationships:

- Should clearly identify relationship participants.
- Should clearly define participation, connectivity, and document cardinality.

ER model:

- Should be validated against expected processes: inserts, updates, and deletions.
- Should evaluate where, when, and how to maintain a history.
- Should not contain redundant relationships except as required (see attributes).
- Should minimize data redundancy to ensure single-place updates.
- Should conform to the minimal data rule: All that is needed is there, and all that is there is needed.

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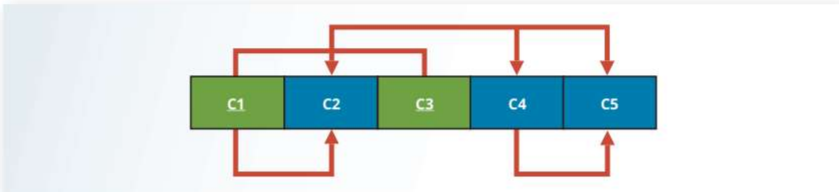
Review Questions

1. What is normalization?
2. When is a table in 1NF?
3. When is a table in 2NF?
4. When is a table in 3NF?
5. When is a table in BCNF?
6. Given the dependency diagram shown in Figure Q6.6, answer Items 6a–6c.

Figure Q6.6 Dependency Diagram for Question 6



a. Identify and discuss each of the indicated dependencies.
b. Create a database whose tables are at least in 2NF, showing the dependency diagrams for each table.
c. Create a database whose tables are at least in 3NF, showing the dependency diagrams for each table.

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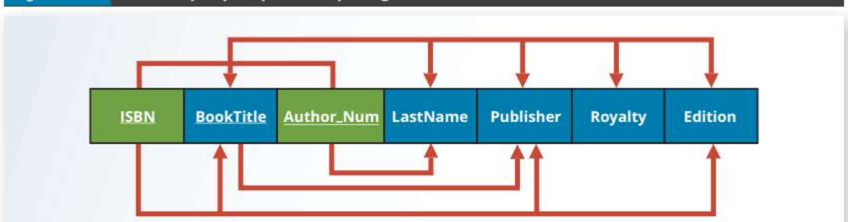
7. The dependency diagram in Figure Q6.7 indicates that authors are paid royalties for each book they write for a publisher. The amount of the royalty can vary by author, by book, and by edition of the book.

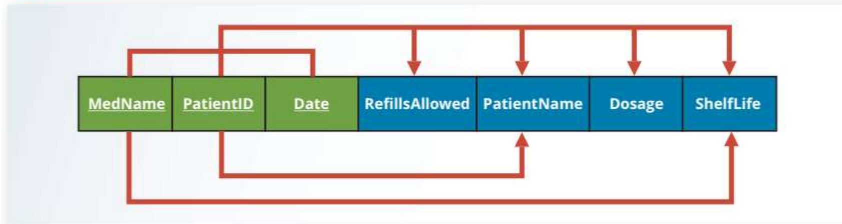
Figure Q6.7 Book Royalty Dependency Diagram



a. Based on the dependency diagram, create a database whose tables are at least in 2NF, showing the dependency diagram for each table.
b. Create a database whose tables are at least in 3NF, showing the dependency diagram for each table.

8. The dependency diagram in Figure Q6.8 indicates that a patient can receive many prescriptions for one or more medicines over time. Based on the dependency diagram, create a database whose tables are in at least 2NF, showing the dependency diagram for each table.

Figure Q6.8 Prescription Dependency Diagram

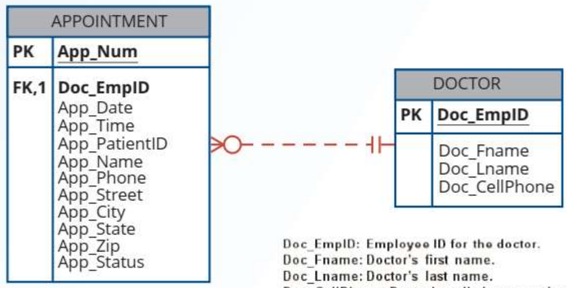


9. What is a partial dependency? With what normal form is it associated?
10. What three data anomalies are likely to be the result of data redundancy? How can such anomalies be eliminated?
11. Define and discuss the concept of transitive dependency.
12. What is a surrogate key, and when should you use one?
13. Why is a table whose primary key consists of a single attribute automatically in 2NF when it is in 1NF?
14. How would you describe a condition in which one attribute is dependent on another attribute when neither attribute is part of the primary key?
15. Suppose someone tells you that an attribute that is part of a composite primary key is also a candidate key. How would you respond to that statement?
16. A table is in _______________ normal form when it is in _______________ and there are no transitive dependencies.

Problems

1. Using the descriptions of the attributes given in the figure, convert the ERD shown in Figure P6.1 into a dependency diagram that is in at least 3NF.

Figure P6.1 Appointment ERD for Problem 1

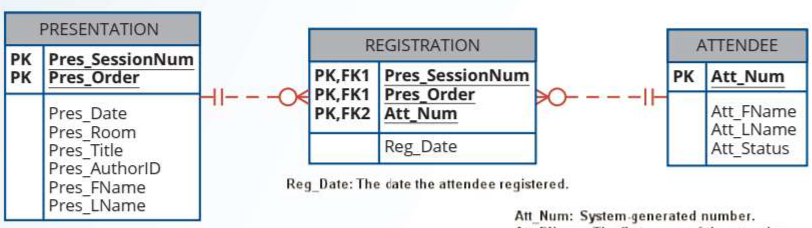


Doc_EmpID: Employee ID for the doctor.
Doc_Fname: Doctor's first name.
Doc_Lname: Doctor's last name.
Doc_CellPhone: Doctor's cell phone number.

App_Num: System-generated number.
App_Date: The date of the appointment.
App_Time: The time of the appointment.
App_PatientID: The ID number of the patient.
App_Name: The name of the patient.
App_Phone: The contact phone number of the patient.
App_Street: The street address for the patient.
App_City: The city the patient lives in.
App_State: The state the patient lives in.
App_Zip: The zip code for the patient's address.
App_Status: The status of the appointment (pending, closed, cancelled)

2. Using the descriptions of the attributes given in the figure, convert the ERD shown in Figure P6.2 into a dependency diagram that is in at least 3NF.

Figure P6.2 Presentation ERD for Problem 2



Reg_Date: The date the attendee registered.

Att_Num: System-generated number.
Att_FName: The first name of the attendee.
Att_LName: The last name of the attendee.
Att_Status: Whether or not the attendee has paid the registration fee.

Pres_SessionNum: System-generated number.
Pres_Order: Number indicating the order of the presentations during the session.
Pres_Date: The date that the presentation is scheduled to be given.
Pres_Room: The room in which the presentation will be given.
Pres_Title: The title of the presentation.
Pres_AuthorID: System-generated number assigned to presentation authors.
Pres_FName: The first name of the presentation author.
Pres_LName: The last name of the presentation author.

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3. Using the INVOICE table structure shown in Table P6.3, do the following:

Table P6.3

Attribute Name	Sample Value	Sample Value	Sample Value	Sample Value	Sample Value
INV_NUM	211347	211347	211347	211348	211349
PROD_NUM	AA-E3422QW	QD-300932X	RU-995748G	AA-E3422QW	GH-778345P
SALE_DATE	15-Jan-2022	15-Jan-2022	15-Jan-2022	15-Jan-2022	16-Jan-2022
PROD_LABEL	Rotary sander	0.25-in. drill bit	Band saw	Rotary sander	Power drill
VEND_CODE	211	211	309	211	157
VEND_NAME	NeverFail, Inc.	NeverFail, Inc.	BeGood, Inc.	NeverFail, Inc.	ToughGo, Inc.
QUANT_SOLD	1	8	1	2	1
PROD_PRICE	$49.95	$3.45	$39.99	$49.95	$87.75

a. Write the relational schema, draw its dependency diagram, and identify all dependencies, including all partial and transitive dependencies. You can assume that the table does not contain repeating groups and that an invoice number references more than one product. (*Hint:* This table uses a composite primary key.)

b. Remove all partial dependencies, write the relational schema, and draw the new dependency diagrams. Identify the normal forms for each table structure you created.